

MT pipeline for FSL

GH, last reviewed 14 Feb 2026

Purpose: Co-registration of 3D MRI to individual space in acpc orientation for calculation of MTsat¹, T1², and B1+³

Data requirements and structure: Dicom 2D images converted to NifTi 3D volumes in arbitrary units [a.u.], that is, of consistent scaling (default on Siemens; rescale GE and Philips) reordered to standard transverse orientation (x=RL, y=AP, z=IS)

File names start with an exam identifier [hum_num] with content identifier appended by “-” for image data or “_” for maps.

Volumes of different angulation and resolution are organized in individual sub-directories to exam.

Target directory /acpc for synopsis of structural and quantitative data.

seg_acpc rigid registration of MP-RAGE (-tfl) and optional SPACE (-tse) from /seg using 6 parameter FLIRT to create an individual reference in acpc angulation in /acpc

mt_acpc rigid registration of MT-weighted (-mt), PD-w (-pd), and T1-w (-t1) from /mt to MP-RAGE in /acpc; brain mask created on -pd using BET, calculation of ‘apparent’ _T1, _A(mplitude); and _MT(sat) maps based on nominal flip angles and TR from masked input (-msk)

bias_acpc calculation of B1+ maps (_bias) from two single-shot STEAM volumes (-tst60) and (-tst110) in /rf, registration to /acpc and correction of _T1 maps to _T1corr. Motion artifacts are mitigated by averaging or exclusion of corrupted volumes

Remark: Data organization and processing strategy had been adapted to the limited functionality of fslview. For clinical application see e.g. ⁴.

1. Helms G, Dathe H, Kallenberg K, Dechent P. High-resolution maps of magnetization transfer with inherent correction for RF inhomogeneity and T1 relaxation obtained from 3D FLASH MRI. *Magnetic Resonance in Medicine*. Dec 2008;60(6):1396-1407.
2. Helms G, Dathe H, Dechent P. Quantitative FLASH MRI at 3T using a rational approximation of the Ernst equation. *Magnetic Resonance in Medicine*. Mar 2008;59(3):667-672.
3. Helms G, Finsterbusch J, Weiskopf N, Dechent P. Rapid radiofrequency field mapping in vivo using single-shot STEAM MRI. *Magnetic Resonance in Medicine*. Sep 2008;60(3):739-743.
4. Dreha-Kulaczewski SF, Brockmann K, Henneke M, Dechent P, Gärtner J, Helms G. Assessment of myelination in hypomyelinating disorders by quantitative MRI. *Journal of Magnetic Resonance Imaging*. 2012;36(6):1329-1338.